FH Salzburg
MultiMediaTechnology

Basics Game Development

Introduction - Lab

Technology
Health
Media

Today's Topics

- Grading
- VSCode & SFML Project Setup
- Draw your first Sprite in SFML (Window, Sprite, View)
- Relevant C# features (Extension Methods, Struct vs Class)
- Debugging in VSCode

Grading

- 3 Assignments that build upon each other
- Assignments themselves will not be graded, but...
- Test after each lesson, containing topics of assignment
- Assignments + slides are the best way to practice for the test
- Assignments will be the basis for your MMP (Game)
- Optionally: Improve your grade (bonus points) by presenting your assignment solution in class (2 slots per exercise, first to mail in advance gets the slot)

- We will be doing a test run of this today, end of lesson

SFML Window

```
static void Main(string[] args)
{
    VideoMode mode = new VideoMode(800, 600);
    RenderWindow window = new RenderWindow(mode, "My window");

    window.Closed += (sender, args) => window.Close();

    while (window.IsOpen)
    {
        window.DispatchEvents();
        window.Clear(Color.Blue);
        window.Display();
    }
}
```

SFML Window - Additions

- window.SetFramerateLimit(60);
- window.SetVerticalSyncEnabled(true);

Offtopic: Lambda-Expressions

- Lambda Expressions are anonymous functions.
- Lambdas are usefull when it comes to window-callbacks in SFML (input, resize, close).
- Statement Lambdas (a certain kind of lambda-expression styles in C#)
  - Syntax: **(**input-parameters**) => {** method-body **}**
- Additional Reading: MSDN

SFML Sprite

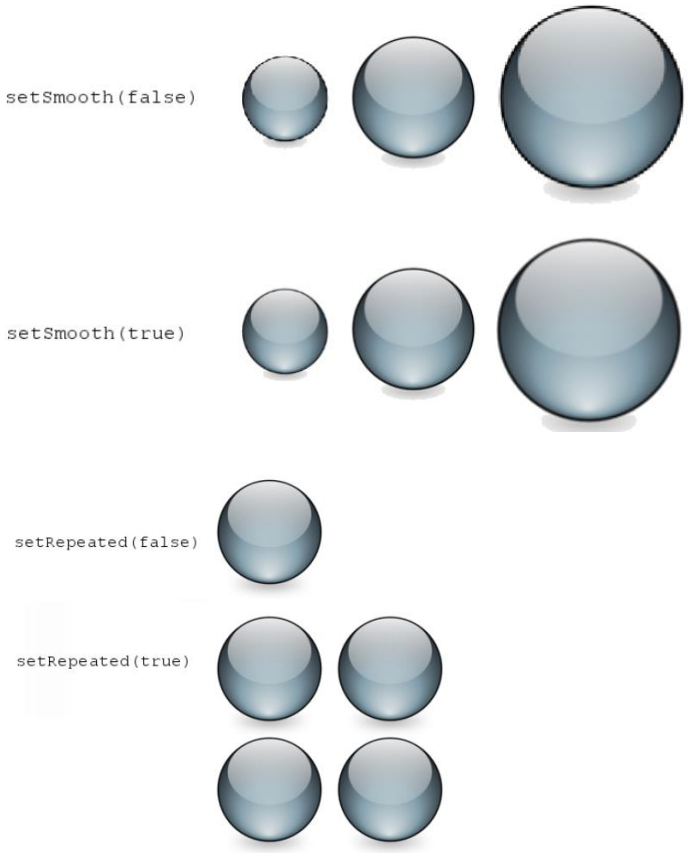
SFML Sprite

```
Texture texture = new Texture("filePath");

Sprite sprite = new Sprite(texture);

sprite.Scale =  new Vector2f(1.0f, 1.0f);
sprite.Position = new Vector2f(0,0);

sprite.TextureRect = new IntRect(0,0, width , height );
```

SFML Sprite Additions

- texture.Smooth = **true**;

- texture.Repeated = **true**;



SFML View

```
// create a View
View view = new View( center, size);

// move a view
view.Move(offset);

// rotate the view at 20 degrees
view.Rotate(20.0f);

// zoom the view relatively to its current size
view.Zoom(0.5f);
```

SFML View - Additions

- Split view (Single Player , MiniMap etc ..)

```
// Split view
view.Viewport = new FloatRect(0.0f, 0.0f, 0.5f, 1.0f);

view02.Viewport = new FloatRect(0.5f, 0.0f, 0.5f, 1.0f);
```

Extension Methods in C#

- Extension methods enable you to "add" methods to existing types without creating a new derived type, recompiling, or otherwise modifying the original type.
- Extension methods are static methods, but they're called as if they were instance methods on the extended type. -> MSDN

```
public static class MyExtensions
{
    public static int WordCount(this string str)
    {
        return str.Split(new char[] { ' ', '.', '?' },
                         StringSplitOptions.RemoveEmptyEntries).Length;
    }
}
```

```
string s = "Hello Extension Methods";
int i = s.WordCount();    👌
```

class vs struct in C#

- A struct is a **value type** that can encapsulate data and related functionality. It is similar to a class, but with some differences:
  - Parameterless constructor must be **public**.
  - Defining a custom constructor will not hide the implicit parameterless constructor.
  - A struct **cannot inherit** from another struct or class, but it can implement interfaces.
  - A struct is **copied by value** when passed to a method or assigned to a variable, while a class is copied by reference.
  - A struct **can't be null**.

See assignment for more details

Example of how to define and use a struct in C#

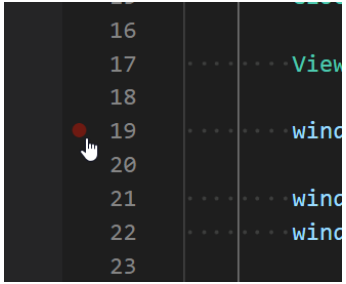
```
// Define a struct
public struct Coords
{
    public Coords(double x, double y)
    {
        X = x;
        Y = y;
    }

    public double X { get; }
    public double Y { get; }

    public override string ToString() => $"({X}, {Y})";
}

// Declare and initialize a struct variable
Coords point = new Coords(10.0, 20.0);

// Access the struct members
Console.WriteLine(point.X); // 10.0
Console.WriteLine(point.Y); // 20.0
Console.WriteLine(point.ToString()); // (10.0, 20.0)
```

VSCode: Helpful Shortcuts

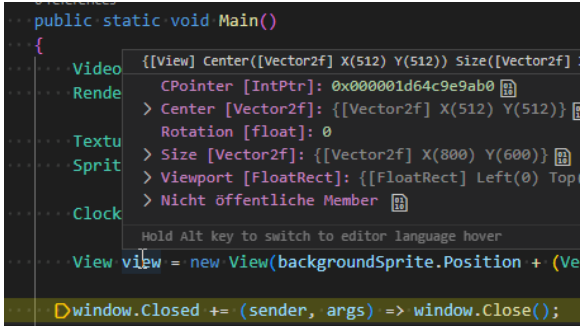
- **Command Palette✨:** F1 or Ctrl + Shift + P
- **Rename:** F2 (updates references)
- **Jump to Definition:** F12
- **Show Usages:** Shift + F12
- **Quick Fix:** Ctrl + .
- **Show Suggestions:** Ctrl + Space (= IntelliSense)
- **Show Parameters:** Ctrl + Shift + Space
- **Run and Attach Debugger:** F5
- **Search for file:** Ctrl + P
- **Format Document:** Shift + Alt + F

Show Shortcuts: Ctrl + K, Ctrl + S

VSCode: Basic Debugging



- Debugging = identify & remove errors from software
- Attach Debugger via VSCode (F5)
- Set Breakpoints by clicking next to line numbers
- Inspect variables by mouse hover / watch
- Helpful Shortcuts:
  - F5: continue execution
  - Shift + F5: stop execution
  - F10: next line
  - F11: step into
  - Shift + F11: step out



VSCode: XML Comments

- Documentation Information when hovering over tokens
- Enabled by XML comments above declaration of token

- Automatically generated by typing '///'

```
VideoMode.VideoMode(uint width, uint height)
width: Video mode width
2/3
Construct the video mode with its width and height
VideoMode(800, 600);
```

```
/// <summary>
/// Construct the video mode with its width and height
/// </summary>
/// <param name="width">Video mode width</param>
/// <param name="height">Video mode height</param>
1 reference
public VideoMode(uint width, uint height)
{
```

Assignment 1
&
MSTeams Quiz – Test Run

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